

UNDERSTANDING SIMPLE MACHINES

DEVICES THAT CHANGE THE SIZE AND DIRECTION OF FORCES HAVE BEEN USED SINCE ANCIENT TIMES

Mechanical devices are composed of different working parts. Among the most important are six basic components called simple machines, which mathematicians and engineers have studied since ancient times: the wheel and axle, the inclined plane, the lever, the pulley, the wedge, and the screw.

Greek engineer Hero of Alexandria (1st century CE) was the first person to bring together the simple machines, in his book *Mechanica*, although the inclined plane was not included in his account. Hero illustrated and explained various devices for lifting heavy objects. Others before him had studied why these devices work—most notably, Archimedes of Syracuse (3rd century BCE), who studied levers. Archimedes worked out that the ratio between the input force (the effort) and the output force (the load) is equal to the ratio between the distances from

the pivot at which those forces act. So, to gain a very large “mechanical advantage” (multiplication of force)—and move a heavy load—a very long lever should be used, but the load needs to be close to the pivot point. What the ancient engineers didn’t realize is that there is always a pay-off between force and the distance—to gain a large mechanical advantage, the long end of the lever moves through a large distance, while the load moves only a small way. Similarly, using pulleys to lift a heavy load, the length of rope you must pull is much greater than the distance the load moves. The amount of “work” done by the effort is the same as the amount of work done by the load (neglecting friction).

HERO
Hero (or Heron) of Alexandria was one of the most prolific engineers of ancient Greece. He is seen here demonstrating his aeolipile, an early example of the use of steam power.

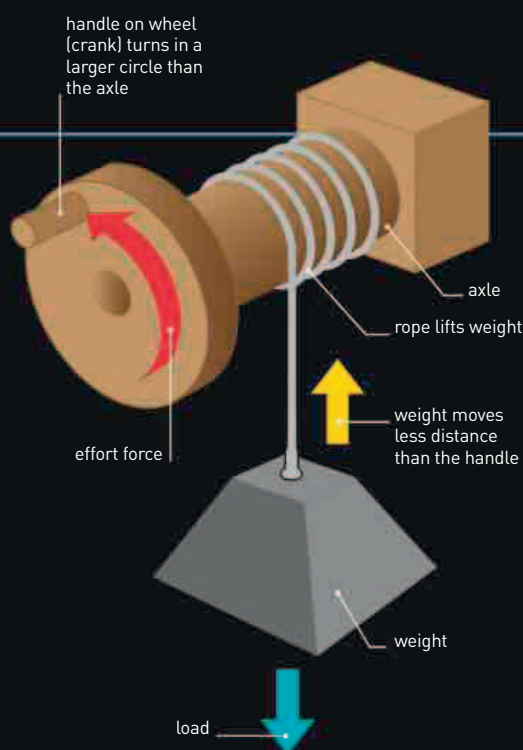


WHEEL AND AXLE

The wheel was invented in Mesopotamia around 3500 BCE. When a wheel is fixed to an axle, the two turn together; ancient engineers used wheels in devices such as the windlass by winding ropes around the axle. The mechanical advantage of a windlass is the ratio of the crank wheel’s radius to the axle’s radius—if the crank wheel has twice the radius of the axle, the effort force will be doubled. Door handles and bicycle cranks are modern examples of the wheel and axle. Gears are interlocking wheels without axles; the mechanical advantage is the ratio of diameters between one gear and the next.

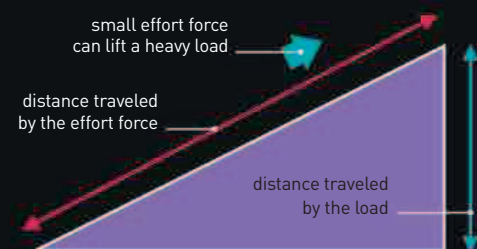
TURNING FORCE

A rope is pulled by an axle turned by a wheel. By making a wheel much larger than the axle, it is possible to gain a large mechanical advantage—but the handle moves through a much greater distance than the weight.



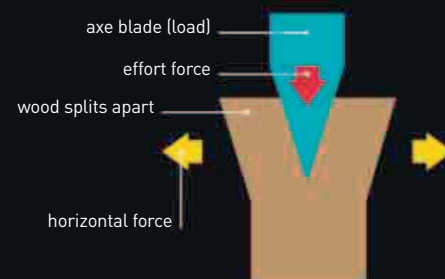
INCLINED PLANE

People have used simple ramps (inclined planes) to gain a mechanical advantage since prehistory. A person raising an object by pushing it up a ramp pushes with a lesser force than if the object were being lifted directly; however, the object must be pushed along the ramp’s length, while the load moves, vertically, a much shorter distance.



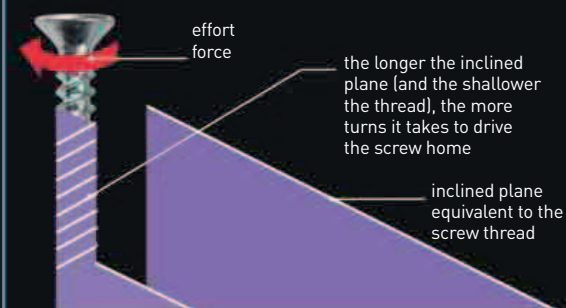
RAMP

The simplest example of an inclined plane is a ramp. A heavy load can be pushed up a ramp in a continuous motion that requires a smaller force than would be required to lift the load straight up.



WEDGE

Two inclined planes back to back make a wedge. An ax blade is a wedge, which, forced vertically into a block of wood, produces a strong horizontal force. The force splits the wood—but the two pieces move only a small distance apart.



SCREW

A screw thread is equivalent to an inclined plane wrapped around a shaft. Turning a screw inside a material pulls it inward. Screws are also used to move water, grain, and other bulk materials in screw conveyors.